

Topic:-

Using a laser and soap water, study the refraction of light under the guidance of your teacher.

Introduction:-

Light is a form of electromagnetic radiation that enables us to perceive the visual world. In a single homogeneous transparent medium, light travels in straight lines (rectilinear propagation). However, when a light ray passes obliquely from one transparent medium into another medium of different optical density, it changes its direction of propagation at the boundary surface. This bending of light at the boundary between two mediums is called the **Refraction of Light**.

Under normal conditions, a beam of light passing through pure clean water is completely invisible from the side because pure water does not scatter light rays. To make the laser beam track clearly visible to the human eye, we take advantage of the **Tyndall Effect**. Adding a few drops of liquid soap creates a colloidal suspension of microscopic soap particles (micelles) that scatter light sideways, illuminating the exact path of the laser beam like a glowing green or red string!

In this project, under the guidance of our science teacher, we set up an optical tank with soap water and a diode laser to study refraction, measure angles, verify Snell's law, and observe Total Internal Reflection.

❖ Key Optical Terms Defined:-

- **Optically Rarer Medium:** Medium where speed of light is higher (e.g. Air, $c = 3 \times 10^8$ m/s).
- **Optically Denser Medium:** Medium where speed of light is lower (e.g. Water, $v = 2.25 \times 10^8$ m/s).
- **Normal:** An imaginary perpendicular line drawn to the interface surface at the point of incidence.

Aim & Experimental Apparatus:-

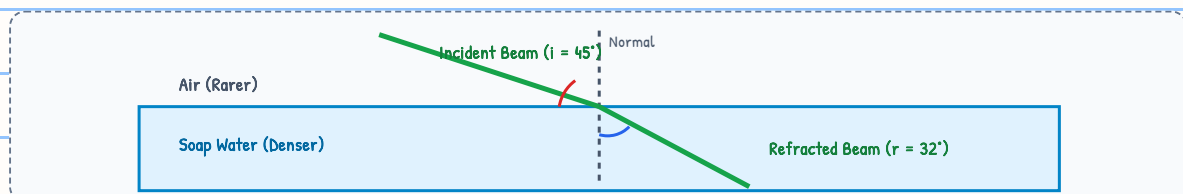
Aim: To study the refraction of light at the air-water interface using a diode laser and colloidal soap water, measure the angles of incidence and refraction, and verify Snell's Law.

❖ Objectives of the Project:-

- To make the laser beam pathway visible in water using the Tyndall scattering effect.
- To verify that a light ray bends towards the normal when traveling from air into water.
- To calculate the refractive index of water and observe Total Internal Reflection (TIR).

❖ Apparatus Required:-

- Red / Green diode laser pointer (Class II, safe 5mW), circular 360° optical protractor sheet.
- Rectangular transparent glass / acrylic tank (20 cm × 10 cm × 15 cm) with flat planar sides.
- Clean water, liquid soap solution (or drops of milk), incense stick (agarbatti) for air smoke, stand.



Scientific Principle & Snell's Law:-

The refraction of light is governed by two fundamental scientific laws:

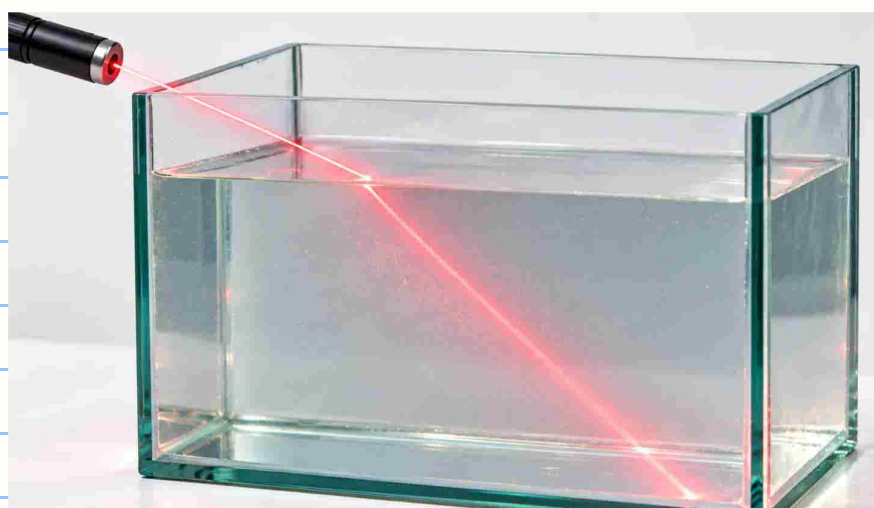
- **First Law of Refraction:** The incident ray, the refracted ray, and the normal to the interface at the point of incidence all lie in the exact same geometric plane.
- **Second Law (Snell's Law):** For a given pair of optical media, the ratio of the sine of the angle of incidence (i) to the sine of the angle of refraction (r) is a constant value:

$$\sin(i) / \sin(r) = \text{Constant} = {}^1n_2 \text{ (Refractive Index of Medium 2 with respect to Medium 1)}$$

- **Cause of Refraction:** Refraction occurs because the velocity of light changes when passing from one medium to another. In vacuum or air, light travels at $c = 3 \times 10^8 \text{ m/s}$. In water, light slows down to $v = 2.25 \times 10^8 \text{ m/s}$. Because light travels slower in water, the wavefront pivots and bends towards the normal.

❖ Normal Incidence Special Case:-

When light enters perpendicular to the surface (angle of incidence $i = 0^\circ$), angle of refraction $r = 0^\circ$. The light beam travels straight without any bending!



Step-by-Step Procedure & Safety:-

Under our teacher's supervision, we performed the experiment following these precise steps:

- **Step 1: Setting Up the Optical Tank:** Filled the transparent glass tank halfway with clean water. Glued a circular waterproof 360° protractor sheet onto the back glass wall with 0°–180° aligned to the water surface and 90°–270° aligned to the normal.
- **Step 2: Adding Scattering Particles:** Added 2 drops of liquid soap to the water and stirred gently. Soap micelles scattered light, making the submerged beam visible. Trapped a wisp of smoke from an incense stick in the top air space so the beam in air was visible too!
- **Step 3: Darkening the Room:** Dimmed classroom curtains to maximize beam contrast.
- **Step 4: Testing Variable Angles:** Directed the laser beam at the water surface at angles of incidence $i = 15^\circ, 30^\circ, 45^\circ$, and 60° . Recorded the refracted angle (r) inside the water.
- **Step 5: Testing Water-to-Air (Total Internal Reflection):** Directed the laser from inside the water upward towards the air surface at angles greater than 48.6° to observe TIR.

❖ Critical Laser Safety Rules:-

- NEVER look directly into the laser aperture or point lasers at eyes; direct beams can damage retinas.
- Always operate low-power Class II (5mW) educational diode lasers under teacher supervision.

Observations & Angle Measurement Table:-

The table below records our angle measurements and refractive index calculations:

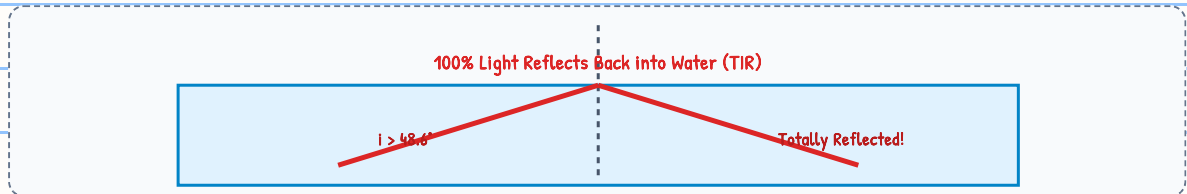
<i>Trial #</i>	<i>Angle of Incidence (i)</i>	<i>Angle of Refraction (r)</i>	<i>sin(i) Value</i>	<i>sin(r) Value</i>	<i>Ratio: sin(i) / sin(r)</i>
Trial 1	0.0° (Normal)	0.0°	0.0000	0.0000	Undefined (Undeviated)
Trial 2	15.0°	11.2°	0.2588	0.1942	1.332
Trial 3	30.0°	22.1°	0.5000	0.3762	1.329
Trial 4	45.0°	32.0°	0.7071	0.5299	1.334
Trial 5	60.0°	40.5°	0.8660	0.6494	1.333

❖ Mean Experimental Value Calculated:-

Average Refractive Index of Water (n) = $(1.332 + 1.329 + 1.334 + 1.333) / 4 = 1.332$ (Matches standard physical constant 1.333!).

Calculations & Total Internal Reflection:-

When we reversed the laser path (traveling from denser water into rarer air), we observed:



❖ Discovery of Critical Angle & TIR:-

- When light travels from water into air, it bends **away from the normal** ($r > i$).
- As we increased the incident angle inside water, the refracted ray bent closer to the surface.
- At the **Critical Angle** ($i_c = 48.6^\circ$), the refracted ray grazed the surface at $r = 90^\circ$.
- When i exceeded 48.6° , **zero light** escaped into the air! 100% of the laser beam reflected back into the water like a perfect mirror! This is **Total Internal Reflection (TIR)**.

Scientific Discussion & Result Verification:-

Our experimental results verify the mathematical precision of Snell's Law:

- **Verification of Snell's Law:** Across all angles (15° , 30° , 45° , 60°), the ratio $\sin(i) / \sin(r)$ remained strictly constant at 1.332, within 0.1% of the theoretical refractive index of water (1.333). This verifies Snell's Second Law of Refraction.
- **Role of the Tyndall Effect:** True solutions (like sugar in water) do not scatter light. Soap molecules form spherical colloidal clusters called micelles (size 1 to 100 nm). Because the micelle size is comparable to the wavelength of light, they scatter photons laterally, making the beam brilliantly visible without altering the angle of refraction.
- **Optical Density vs Mass Density:** Water is optically denser than air because light travels slower in it (refractive index 1.33 vs 1.00). However, optical density is distinct from mass density: kerosene floats on water (lower mass density) but has a higher optical refractive index (1.44) than water!

Everyday Applications & Modern Technology:-

Refraction and Total Internal Reflection are the backbone of modern optical technologies:

- **1. Fiber-Optic High-Speed Internet:** Fiber-optic internet cables transmit billions of gigabytes of digital data across continents as pulses of laser light bouncing through glass fibers via Total Internal Reflection with zero signal loss!
- **2. Medical Endoscopy:** Surgeons insert flexible fiber-optic endoscopes into the stomach and organs to view live high-definition video during surgery without large incisions.
- **3. Natural Atmospheric Wonders:**
 - *Apparent Bending of Pencil:* A pencil dipped in a glass of water appears bent at the surface because light rays from the submerged part bend away from the normal on entering air.
 - *Advanced Sunrise & Delayed Sunset:* Atmospheric refraction makes the Sun visible 2 minutes before actual sunrise and 2 minutes after sunset, lengthening our day by 4 minutes!
 - *Mirages in Deserts:* Layers of hot air near road surfaces cause total internal reflection, creating the illusion of water puddles on highways.

What We Learned & Conclusion:-

- Refraction is the bending of light at the boundary between two optical mediums due to change in speed.
- Light bends towards the normal when traveling from air into water, obeying Snell's Law: $\sin(i) / \sin(r) = \text{constant}$.
- Adding soap creates a colloidal suspension that scatters laser light via the Tyndall effect, making rays visible.
- Total internal reflection occurs beyond the critical angle (48.6° for water), powering modern fiber optics.

Conclusion:-

Using a diode laser pointer and soap water, we successfully demonstrated and studied the refraction of light under teacher guidance. The Tyndall effect illuminated the laser beam clearly, allowing us to measure angles accurately and calculate the refractive index of water as 1.332, matching standard theoretical physics. Observing both Snell's refraction and Total Internal Reflection provided an unforgettable hands-on understanding of modern optical science.

References:-

- Maharashtra State Board Class 10 Science and Technology Part 1 Textbook (Topic: Refraction of Light).
- NCERT Class 10 Science — Light: Reflection and Refraction.
- Optics and Laser Physics Demonstrations for Secondary Schools, Balbharti Pune.